

Neither Reverse Extremal Norm Envelope Need Exist

Full counterexample packet for arXiv:1604.02812
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Status: candidate counterexample (full answer). On $\mathbb{M}_3$, an explicit norm has no greatest L -norm below it. The dual norm has no least M -norm above it. Thus both general existence questions in Kian have negative answers already in dimension three.

1 Source question

Kian [1, p. 11, after Corollary 3.5] proves the existence of the opposite extremal envelopes and asks:

1. Does the class of L -norms below an arbitrary norm contain a greatest element?
2. Does the class of M -norms above an arbitrary norm contain a least element?

The source answers these only in special cases. We answer both questions in the negative.

Corollary 3.5. *The operator norm $\|\cdot\|_\infty$ is the maximum element in the class of M -norms smaller than $\omega_*(\cdot)$, the dual norm of $\omega(\cdot)$.*

Proof. Apply Corollary 3.2 and Theorem 3.4. $\square$

A natural question is that does the class of L -norms $\|\cdot\|$ which are less than a norm $\|\cdot\|$ contain a maximum element? or does the class of M -norms $\|\cdot\|$ which are greater than a norm $\|\cdot\|$ contain a minimum element?

We answer these questions in some special cases below. First we need the following simple lemma.

Lemma 3.6. *If P and Q are orthoprojections in $\mathbb{B}(\mathcal{H})$, then*

$$\|PXP + QYQ\|_1 = \|PXP\|_1 + \|QYQ\|_1$$

for every $X, Y \in \mathbb{L}^1(\mathcal{H})$.

Proof. First note that $\text{Tr}(U^*AU) = \text{Tr}(A)$ for every $A \in \mathbb{L}^1(\mathcal{H})$ and every unitary U . If P and Q are orthoprojections in $\mathbb{B}(\mathcal{H})$, then $U = \begin{bmatrix} P & Q \\ Q & P \end{bmatrix}$ is a unitary in $\mathbb{B}(\mathcal{H} \oplus \mathcal{H})$. If $A, B \in \mathbb{L}^1(\mathcal{H})$, then

$$\begin{aligned} \text{Tr}(PAP + QBQ) + \text{Tr}(QAY + PBP) &= \text{Tr} \left(\begin{bmatrix} PAP + QBQ & PAQ + QBP \\ QAP + PBQ & QAY + PBP \end{bmatrix} \right) \\ &= \text{Tr} \left(\begin{bmatrix} P & Q \\ Q & P \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} P & Q \\ Q & P \end{bmatrix} \right) \\ &= \text{Tr} \left(\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \right) = \text{Tr}(A) + \text{Tr}(B). \end{aligned} \quad (14)$$

Put $A = (PX^*PXP)^{\frac{1}{2}} = |PXP|$ and $B = (QY^*QYQ)^{\frac{1}{2}} = |QYQ|$. Then

$$\begin{aligned} \text{Tr}(PAP + QBQ) &= \text{Tr} \left(P(PX^*PXP)^{\frac{1}{2}}P + Q(QY^*QYQ)^{\frac{1}{2}}Q \right) \\ &\leq \text{Tr} (PX^*PXP + QY^*QYQ)^{\frac{1}{2}} \quad (\text{by the Jensen inequality}) \\ &= \text{Tr} \left(((PXP + QYQ)^*(PXP + QYQ))^{\frac{1}{2}} \right) \\ &\quad (\text{by orthogonality of } P \text{ and } Q) \\ &= \text{Tr}(|PXP + QYQ|) = \|PXP + QYQ\|_1. \end{aligned} \quad (15)$$

Moreover,

$$\begin{aligned} \text{Tr}(QAY + PBP) &= \text{Tr} \left(Q(PX^*PXP)^{\frac{1}{2}}Q + P(QY^*QYQ)^{\frac{1}{2}}P \right) \\ &\leq \text{Tr} (QPX^*PXPQ + PQY^*QYQP)^{\frac{1}{2}} \quad (\text{by the Jensen inequality}) \\ &= 0. \end{aligned} \quad (16)$$

Figure 1: Source PDF page 11. The two general existence questions appear immediately after Corollary 3.5.

2 Preliminaries

Let ω denote the numerical-radius norm and ω_* its dual under the trace pairing $\langle A, B \rangle = \text{Tr}(A^*B)$. We use two elementary facts from the source framework:

- the trace norm $\|\cdot\|_1$ and ω_* are L -norms;
- duality reverses norm order and takes L -norms to M -norms.

For completeness, the first fact follows because the operator norm and numerical radius are M -norms. For the numerical radius this is immediate: if $\sum C_i^* C_i = I$, then

$$\omega\left(\sum_i C_i^* A_i C_i\right) \leq \max_i \omega(A_i).$$

Indeed, apply the quadratic forms to a unit vector and use $\sum_i \|C_i x\|^2 = 1$. The operator-norm statement follows by writing the sum as a compression of the block-diagonal operator $\text{diag}(A_1, \dots, A_k)$.

We will also use the following order-theoretic observation.

Lemma 1. *Let a and b be L -norms, and suppose $p = \max\{a, b\}$ is not an L -norm. Then there is no greatest L -norm below p .*

Proof. Both a and b lie below p . If a greatest such L -norm g existed, then $g \geq a, b$, hence $g \geq p$. Since also $g \leq p$, one would have $g = p$, contradicting the hypothesis. $\square$

3 The explicit counterexample

Let $E = E_{11}$, $J = E_{23}$, and $X = E + J$ in $\mathbb{M}_3$. Thus, relative to $\mathbb{C} \oplus \mathbb{C}^2$,

$$X = 1 \oplus \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Put

$$p(A) = \max \left\{ \|A\|_1, \frac{3}{4} \omega_*(A) \right\}. \quad (1)$$

This is a norm and is the maximum of the two L -norms $a = \|\cdot\|_1$ and $b = \frac{3}{4} \omega_*$.

The relevant values are

A	E	J	X
$\ A\ _1$	1	1	2
$\omega_*(A)$	1	2	3
$p(A)$	1	$\frac{3}{2}$	$\frac{9}{4}$.

(2)

Here is a short exact verification of the second row. Clearly $\omega_*(E) = 1$. Since $\omega(J) = 1/2$, the test matrix $2J$ gives $\omega_*(J) \geq 2$; the reverse inequality follows from the standard bound $\|Y\|_\infty \leq 2\omega(Y)$. Finally, the triangle inequality gives $\omega_*(X) \leq 3$, while the block-diagonal test matrix $E + 2J$ has numerical radius one and

$$\text{Tr}(X^*(E + 2J)) = 3,$$

so equality holds.

Theorem 2. *The norm p in (1) has no greatest L -norm below it.*

Proof. Let $P = E$ and $Q = I - E$. These are orthogonal projections with $P^*P + Q^*Q = I$, and

$$PXP = E, \quad QXQ = J.$$

The L -norm inequality for p would require

$$p(PXP) + p(QXQ) \leq p(X).$$

But (2) gives

$$1 + \frac{3}{2} = \frac{5}{2} > \frac{9}{4}.$$

Thus p is not an L -norm. Lemma 1, applied to $a = \|\cdot\|_1$ and $b = \frac{3}{4}\omega_*$, finishes the proof. $\square$

Robustness. The same construction works with $\frac{3}{4}$ replaced by every $c \in (1/2, 1)$. Indeed, for $p_c = \max\{\|\cdot\|_1, c\omega_*\}$ one has

$$p_c(E) + p_c(J) = 1 + 2c > \max\{2, 3c\} = p_c(X).$$

4 The dual counterexample

Let $r = p_*$. It can be written explicitly as the infimal convolution

$$r(A) = \inf_{A=B+C} \left(\|B\|_\infty + \frac{4}{3}\omega(C) \right). \quad (3)$$

Indeed, the unit ball of the dual of a maximum is the convex hull of the two dual unit balls.

Theorem 3. *There is no least M -norm above the norm r in (3).*

Proof. The two M -norms

$$q_1 = \|\cdot\|_\infty = a_*, \quad q_2 = \frac{4}{3}\omega = b_*$$

both dominate $r = p_*$. If a least M -norm $q \geq r$ existed, then $q \leq q_1, q_2$. Dualizing these three inequalities would give an L -norm q_* satisfying

$$a \leq q_*, \quad b \leq q_*, \quad q_* \leq r_* = p.$$

Hence $q_* = \max\{a, b\} = p$, contradicting Theorem 2. $\square$

5 Scope and provenance

Theorems 2 and 3 settle both source questions as universally stated. They do not contradict the positive special cases in the source: extremal reverse envelopes can exist for particular reference norms, but need not exist for arbitrary ones.

A bounded literature search using the source identifier and title, the exact “maximum L -norm” and “minimum M -norm” phrases, and the relevant norm keywords found no later paper giving this counterexample or an explicit full resolution. This packet is therefore presented as a new candidate counterexample for human review.

References

- [1] M. Kian, *C*-convexity of norm unit balls*, J. Math. Anal. Appl. 445 (2017), 1417–1427; arXiv:1604.02812, doi:10.1016/j.jmaa.2016.04.022.